

Watching Live: The Master Checklist and Anticipation

Twenty seconds between snaps, nine questions in the right order, the full catalogue of tells, and a prediction with a number on it.

The NFC Championship Game, January 25, 2026. Third quarter, 8:23 to play, and the Rams have just pulled within four points. Seattle faces third-and-9 at its own 36, and on a typical broadcast the replay of the run that just lost two yards would still be rolling as the Seahawks come back to the line: one running back, one tight end, three receivers, quarterback in the shotgun. The Rams answer with six defensive backs and only five men near the ball. You have about twenty seconds. Most viewers spend them waiting for the snap. A coach in the booth spends them asking a set of questions in the same order he asks them before every snap: *What does the situation demand? Who's on the field? How are they lined up? What does the defense do when someone moves? How many are in the box? How many are deep? Where is the help? Is anyone coming?* And then he says out loud what he thinks will happen, and how sure he is. This chapter puts the whole course into that routine. By the end of it you'll run the same questions in the same order, and you'll keep score.

You'll be able to...

- Run the master pre-snap checklist in about twenty seconds, in order: situation → personnel → formation and strength → motion response → box → shell → leverage → pressure tells → prediction, and know which steps to keep when the offense goes fast.
- Use the full tells catalogue (linemen's stance and weight, the back's depth and offset, pistol against shotgun, receiver splits, the tight end's stance, linebacker depth, defensive back depth and leverage, safety width), saying for each what it usually means and why it leaks, and, where public data can measure it, how often it lies.
- Say what each step of the checklist is worth, from 2024–2025 data, and why the later steps answer different questions from run or pass.
- Make layered predictions (run or pass first, then the coverage family, the rush, where the ball goes or which way the run hits), each with a stated confidence between 50% and 90%.
- Follow a complete playoff drive snap by snap the way a coach in the booth does, and keep the log that Film Study will score.

You'll need

This is a Level 5 chapter: it assembles the course rather than adding new schemes. One line on each piece it leans on hardest:

- **In-game adjustments and tempo** (In-Game Adjustments, Tempo, and the Sideline Battle): tempo is how fast the offense gets to the line and snaps; a hurry-up offense snaps as fast as it can, and the substitution freeze means a defense that can't match a substitution has to play with the eleven it has. Staffs adjust between series, and the play that follows a successful call (the call after the call) often builds on it.
- **The clock and fourth down** (The Clock, the Two-Minute Drill, and Fourth Down): in the two-minute drill and the four-minute offense the clock decides much of the play call, and fourth down is a decision of its own.
- **Tendencies from data** (Tendencies: PROE, Personnel, Motion, and Coverage from Charting Data): xpass (expected pass rate) is the chance of a pass given the situation alone, and PROE (pass rate over expected) is how much a team throws beyond it. A team's tendency is a pattern in what it calls from a given situation or look.
- **The habit** (How to Watch a Broadcast Like You Mean It): a tell is a pre-snap clue that shifts the odds, never a guarantee; the prediction log is your record of calls, with a confidence column you've been filling in since Part 1. The "checklist so far" cards at the end of Motion and Shifts, Offensive Systems, Disguise and Rotation and Zone Blitzes are the drafts of the card this chapter finishes.

Twenty seconds, a hundred clues

Start with the problem the checklist solves: there's far more to see before a snap than anyone can see, and very little time. The play clock is 40 seconds, starting the moment the previous play is whistled dead (25 seconds after some stoppages, such as a timeout

or a change of possession).¹ On a broadcast the first part of that is usually spent on replays of the play that just ended. The offense huddles while its play-caller talks to the quarterback through the helmet radio, which goes silent with 15 seconds left on the play clock.² Then the players walk to the line, set, maybe move someone in motion, and the ball is snapped. A team that huddles spends most of what is left of the play clock at the line; a team in tempo can snap within a few seconds of setting.

So your budget is roughly twenty seconds, from the moment the scorebug updates to the snap, and the camera won't show you all of it. Within those twenty seconds, information doesn't arrive all at once. It arrives in a fixed order, because the game produces it in a fixed order:

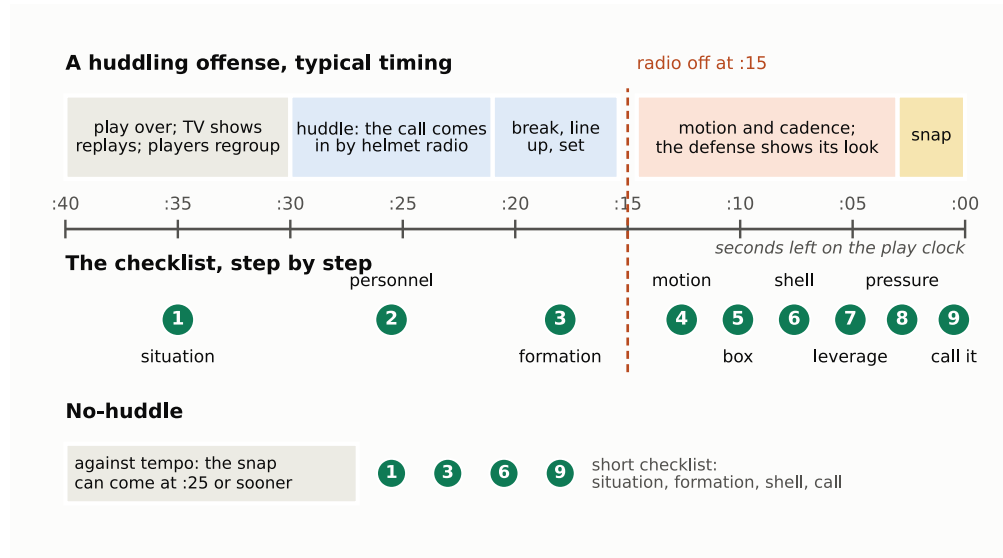


Figure 1: Twenty seconds, in order. Top: what a typical huddling offense does with the 40-second play clock (timings vary by team and snap; only the radio cut-off at :15 is a rule). Middle: the nine pre-snap steps of the master checklist, placed where each answer usually becomes visible: the situation during the replay, the personnel as players run on, the formation when the offense sets, the motion and the defense's reply during the cadence, and the box, shell, leverage and pressure in the last few seconds, when the defense finally shows its hand. Bottom: against tempo everything is compressed, so you keep only steps 1, 3, 6 and 9.

That ordering is the first reason the checklist has the shape it has: **each step is asked when its answer is visible**. There is no point hunting for the safeties while the offense is still in the huddle; they haven't settled. There is every point in reading the scorebug during the replay, because the situation is already fixed and nothing else is.

The second reason is less obvious and more important: **each answer changes what the next clue means**. A safety walking down to eight yards on third-and-9 is a defense guarding the sticks; the same move on first-and-10 against two tight ends is an eighth man for the run. A receiver tightening his split means one thing from a team that runs on 70% of its snaps in that look and another from a team that throws on 70%. Clues aren't independent bits of evidence to be added up. They are answers to questions, and the questions come in an order. Read the box before you know the personnel, and you'll miscount it. Read the shell before you know the down, and you won't know what you're looking at.

¹NFL Football Operations, "Football terms" glossary (the 40-second play clock, and the 25-second clock after certain administrative stoppages): <https://operations.nfl.com/learn-the-game/nfl-basics/terms-glossary/>

²NFL, *Official Playing Rules*, Rule 5, Section 3, Article 3 ("Speakers in helmets"): coach-to-player communication "is cut off when the play clock reaches 15 seconds or the ball is snapped, whichever occurs first." Rulebook: <https://operations.nfl.com/the-rules/nfl-rulebook/>. The rule and its history are covered in The Pre-Snap Phase.

! Why it exists: a checklist, of all things

A checklist sounds like the opposite of expertise. Surely the expert just *sees* it? But the professions that have to perform under time pressure without missing anything (pilots running a pre-flight check, surgical teams before an incision) settled long ago on short, ordered checklists, and not because their people were beginners. A checklist offloads the *remembering* so that attention is free for the *judging*.³ Yours is the kind Gawande calls *do-confirm*: you run it from memory, then glance at the card to make sure you skipped nothing. The football version works the same way. The quarterback's pre-snap routine (count the safeties, find the Mike, know where the hot throw goes) is a checklist; so is the defensive signal-caller's (formation, strength, the check that goes with it). Your job is easier than theirs, since nobody is about to hit you. But it has the same enemy: the eye goes to the most interesting thing on the screen (usually the quarterback) and stays there, and the clue that decided the play happens forty yards away.

What it gives up: a fixed routine is slower than a glance, and on the snaps where the offense hurries you'll have to cut it short. That's why the card has a short version.

The master checklist

Here is the card you've been building since Part 2, finished. It is the course's **master pre-snap checklist**: the complete pre-snap routine, in the same order before every snap, followed by what to check at the snap and between snaps.

³Atul Gawande, *The Checklist Manifesto: How to Get Things Right* (Metropolitan Books, 2009), on aviation checklists, the surgical safety checklist, and "do-confirm" against "read-do" checklists.

The master checklist

new in the capstone

BEFORE THE SNAP • about 20 seconds

- 1 3 s Situation: down, distance, field zone, hash (which side is the field), clock, score, timeouts. Say the base rate: 1st-and-10 about 50%, 3rd-and-7 to 10 about 94%, empty about 95% pass. Two-minute or four-minute? Broadcast · Clock · Tendencies
- 2 2 s Personnel: count backs and tight ends, and notice which receivers. Did the defense match (base, nickel, dime), or did tempo freeze it? Personnel · Tempo
- 3 3 s Formation and strength (to the tight end, the receivers or the field?): under center, gun or pistol; the back's depth and offset; splits; the tight end's stance; heavy or light hands; a guard cheating back Formations · tells
- 4 2 s Motion response: does a defender travel (man), bump (zone) or spin a safety? A lean, not a read. Still moving at the snap? Motion · Disguise
- 5 2 s Box: blockers against defenders near the line. Light (6 or fewer) or loaded (8 or more)? Box math
- 6 2 s Shell: one-high or two-high; each safety's depth and width; who is creeping late? Coverage · Disguise
- 7 2 s Leverage: corners press or off, inside or outside; linebacker depth; where is the help? Coverage · tells
- 8 2 s Pressure tells: mugs, a safety or nickel walked up, a lineman on his heels; the center or QB pointing, a slide call. Count the deep players; where is the hot? Pressure · tells
- 9 2 s Call it: run or pass with a stated confidence (50–90%), plus one second-layer call (coverage, rush, target, or the run's direction) this chapter

AT THE SNAP

- 10 Linemen's first step (run or pass; together = zone, a puller = gap); did a safety rotate; how many rushed? Run game · Pressure

ONE SECOND AFTER

- 11 Middle of the field closed or open; corners squat, bail or trail; where did the ball go? Coverage · Adjustments

BETWEEN SNAPS

- 12 Log it: call, confidence, the clue that decided it, the result. Five seconds, every snap Broadcast · this chapter
- 13 At each stoppage: update today's tendencies (this personnel, this down); tempo and clock status; what did they just adjust? On fourth down, call go, kick or punt Adjustments · Clock

Figure 2: The master checklist: the cards from Parts 2, 4, 6 and 7, completed and reordered for speed. Nine pre-snap steps take about twenty seconds in all (the seconds column is a target, not a rule). The shaded lines are what this chapter adds: the full tells catalogue folded into steps 3, 7 and 8, the prediction with a stated confidence as an explicit step, and the between-snaps habits that keep your read of this team, today, up to date. The right-hand column names the part of the course where each line was taught.

Compare it with the last draft, the seventeen-line card at the end of Zone Blitzes and Simulated Pressure, and three changes stand out. **It's shorter before the snap**: nine steps, not ten, because the protection question (where is the back, which way will the line slide) has been folded into the pressure step, where you need it. **The tells have a home**: instead of a separate “run tells” and

“pass tells” line, each tell sits in the step where you’d see it, offensive alignment tells in step 3, defensive depth and leverage in step 7, pressure tells in step 8. And **the prediction is a step of its own, with a number**: not “I had a feeling it was a pass” after the fact, but “pass, 80%” before it.

The rest of this section walks the nine pre-snap steps in order. For each: what you’re looking at, what it tells you, and why it sits where it does. None of it is new; what’s new is the order and the speed.

Step 1: situation, the base rate

Down, distance, field zone, clock, score, timeouts. You learned to read the scorebug in five seconds in How to Watch a Broadcast, and the situational chapters (Part 10, from Third Down to The Clock) taught you what each situation does to the play-caller. The output of step 1 is a number: the **base rate**, how often offenses throw in this situation before anyone has lined up. That is what xpass measures, and it’s the anchor every later step moves.

Add one more thing a coach reads first: **where the ball is spotted**. On a hash, one side of the formation has far more room (the field side) than the other (the boundary side). Defenses set their strength, rotations and blitzes by it, and offenses put their formations and perimeter plays into the room: in 2025, screens thrown from a hash went to the wide side 63% of the time.⁴

You don’t need a model in your head, just a few landmarks from the 2025 regular season:⁵

Situation	Pass rate, 2025	What it means for your call
All snaps	60%	“Always pass” is the floor any reader has to beat
1st-and-10	50%	A coin flip until the formation speaks
2nd-and-1 or 2	31%	Mostly runs, but this is where shots happen
3rd-and-1 or 2, under center	19%	Run, and the formation says so
3rd-and-1 or 2, shotgun	52%	A genuine coin flip
3rd-and-7 to 10	94%	Pass; spend your seconds on the second layer
No-huddle, any down	70%	Mostly offenses that are behind; under one in five comes in a two-minute drill

Two things to say alongside the number. First, **is this a clock situation?** In the two-minute drill and the four-minute offense the clock calls half the plays, and The Clock gave you a separate set of questions for them (in bounds or out, timeout or spike, kneel or not). Second, **does this team break the base rate?** That is PROE. Before kickoff, copy three numbers per team from the one-page scouting report in Tendencies onto your sheet: its PROE, its pass rate from under center, and its play-action rate. A team with a high PROE moves your 50% first down toward 55 or 60 before you’ve seen a thing.

Why it comes first: it’s the only clue that’s complete during the replay, and it is the biggest single clue there is. Everything else adjusts it.

Step 2: personnel, and the match

Count the backs and tight ends as the offense runs on, and watch who runs *off*. The two digits (11, 12, 21, 13) were the first thing Personnel Groupings taught, because personnel is chosen before the formation: the play-caller sends in a group, and the group limits what the formation can be. Then notice *which* players came on, not only how many. Defenses substitute for players as well as for numbers: a star receiver on the field can draw a nickel even when the digits say two tight ends (you’ll see exactly that on the drive below). Then glance at the defense’s answer. Did it stay in base, go to nickel, go to dime? A defense in nickel against 12 personnel has a defensive back where a linebacker usually stands, and the offense will often try to run at him; a defense in base against 11 personnel has a linebacker who may have to cover a slot receiver, and the offense will often try to throw at him.

⁴Course calculation from nflverse play-by-play ([pass_location](#), [run_location](#)) and FTN charting ([starting_hash](#), [is_screen_pass](#)), 2025 regular season, plays snapped from a hash that went left or right. FTN does not say whose left and right it means; the screens are the evidence that it is the offense’s, because a screen is the play most often sent to the wide side, and from FTN’s “L” hash most screens go right. The same check shows that runs from a hash split almost evenly (48% to the field), so the hash alone doesn’t call a run’s direction.

⁵Course calculation from nflverse play-by-play joined to FTN charting (quarterback alignment, tempo) and participation (personnel), 2025 regular season, runs and dropbacks with a down recorded. “Pass” is nflverse’s [pass](#) flag, so sacks and scrambles count as passes, as in your log.

This is also where you notice **tempo**. If the offense hurries to the line without substituting, the defense may be stuck with the eleven it has, the substitution freeze from In-Game Adjustments and Tempo. An offense that goes fast right after a big gain, or right after the defense has subbed into a run-stopping group, is usually hunting a mismatch it has already spotted.

Why second: personnel is visible before the formation is, and it's the context for everything the formation shows. As you'll see shortly, it adds surprisingly little to a run-or-pass call on its own. Its value is in what it sets up.

Step 3: formation, strength and the offensive tells

This is the richest step. Where is the quarterback: under center, in the shotgun, in the pistol? Where is the back, how deep, offset to which side? Where is the tight end, on the line or off it, in a three-point stance or standing up? How wide are the receivers? Which side is the formation strength?

Strength matters because both sides organise around it. The defense sets its front and usually its safety rotation to the strong side, and the offense's runs and its best route combinations usually go there too. But "strong" is a dialect, not a fact: a defense may declare it to the tight end, to the side with more receivers, or to the field, and its choice decides which way the front and the rotation set (Formation Rules and Vocabulary showed all three). When the tight end is on one side and three receivers are on the other, watch which way the linebackers shift; that tells you which dialect this defense speaks.

The vocabulary came from Formation Rules and Vocabulary, the families of plays each look goes with from the run-game and passing-game parts, and the tells catalogue below collects every alignment clue in one place.

Why third: the offense has to show its formation before the defense commits to its final picture, and the defense's last-second picture is a *response* to it. Read the formation first and you'll understand what the defense does next.

Step 4: motion response

When a player moves before the snap, watch the defense, not the man in motion. A defender who travels across the formation with him suggests man coverage; defenders who bump a spot over suggest zone; a safety who spins down turns two-high into one-high. That's the man/zone indicator from Motion and Shifts as Weapons. Then ask the offensive question: is he still moving when the ball is snapped (motion at the snap)? A jet motion at full speed is a run threat, or a decoy for one, to that side.

You will get to use this step on most snaps. PFF's charting had a shift or motion on 63.9% of 2025 snaps, the highest in its series, and every team but the Giants used one on at least half its plays.⁶

Treat the answer as a lean, not a read. Defenses know this indicator as well as you do: a man defense can bump over and swap assignments (an exchange), and a pattern-matching zone can send a defender across with the motion, which is how Motion and Shifts showed the indicator lying. File the motion response next to leverage and the shell; when all three agree, believe it.

Why fourth: motion only happens after the offense sets, and its whole point is to make the defense answer a question before the snap. The answer is a clue for steps 5 to 7.

Step 5: the box

Count the defenders in the box and compare them with the blockers (box count). Six blockers against six defenders is a fair fight for a run; six against eight means two defenders will be unblocked, which is why an offense that sees eight usually throws or checks to a run away from the extra man (Run Fits and Box Math). A light box (six or fewer) against a shotgun offense is the classic invitation for a run or an RPO.

Why fifth: the box isn't final until motion has settled it. Count too early and you'll count a safety who is about to leave, or miss one who is about to arrive.

Step 6: the shell

One-high or two-high? Then the details: each safety's depth (13 yards is deep, 8 is down), his width (on the hash or inside it, or out wide), and any creep in the last seconds of the play clock. This was the spine of the coverage chapters, and Disguise, Rotation, and Split-Field Coverage was about why the shell you see is often not the coverage you get. The numbers show the gap. After the snap, NFL Next Gen Stats measured two-high coverage on 42.0% of dropbacks in 2025, the highest since it began tracking in

⁶Daire Carragher, "NFL pre-snap motion usage reaches new highs across the NFL," PFF (Feb. 23, 2026): <https://www.pff.com/news/nfl-pre-snap-motion-usage-reaches-new-highs-across-the-nfl>. PFF's series counts shifts and motion together and is not comparable with FTN's *is_motion* flag, which nflverse defines as motion "before or at the time of the snap."

2018.⁷ Before the snap, defenses show two-high far more often than that: preparing for Seattle's defense before Super Bowl LX, the Patriots' offensive coordinator, Josh McDaniels, said 65 of 70 snaps would "look the same before the ball's snapped"⁸ (Fangio's Two-High Revolution). The distance between the shell shown and the coverage played is the disguise. So two-high before the snap is not a tell by itself; the tell is which of the two safeties is shallower, narrower or moving.

Why sixth: safeties are the last defenders to settle and the most likely to be lying, so you read them as late as you can.

Step 7: leverage and depth

Look at the corners: pressed at the line or off at seven yards; on the receiver's inside shoulder or his outside. Leverage tells you what the defender is protecting and so where his help is: a corner playing outside leverage is giving up the inside because someone (a safety in the middle, a linebacker underneath) is there to take it. Then the linebackers' depth: at four and a half to five and a half yards they are playing the run and the short middle; creeping to three, they may be coming; backing out to eight before the snap on a passing down, they are getting a head start into zone drops.

Why seventh: leverage is the last thing a defense adjusts, and it only makes sense after you know the shell. An inside-leverage corner with two deep safeties means something different from the same corner with one.

Step 8: pressure tells

How many defenders are at or near the line, and could any of them drop? Linebackers mugging the A gaps, a safety or nickel back walked up beside the end, a defensive lineman sitting back on his heels: these are the pressure look tells from the pressure chapters. Then the count that caps everything: **how many players are deep?** Two deep safeties and two off corners leave seven to rush and cover underneath, and a defense almost never rushes six with only one underneath. Six-man pressure is real, but it comes as Cover 0, man-to-man with nobody deep, and that needs the safeties to walk down first; if they're still at 12 yards at the snap, six isn't coming.

Finish with the offense's side, which is the easiest pressure tell to see on a broadcast: **the offense reacting**. The center pointing and calling out the middle linebacker, the quarterback stepping up to reset the protection or turning to the sideline, a back sliding across to the other side of the quarterback: each says the offense sees a pressure and has changed who blocks whom. Then ask which way the line will slide and where the hot route goes.

Why eighth: pressure looks are deliberately last-second. Defenses walk up late so the offense has as little time as possible to change its protection, and you get the same late look the quarterback does.

Step 9: call it, with a number

Say it out loud before the snap: run or pass, and how sure you are. Then, if you have a second left, one more layer: the coverage family, how many will rush, or where the ball should go. This is the step this chapter adds, and it gets its own section below. The rule is simple: the call is made *before* the snap, or it doesn't count.

What each step is worth

How much does each step actually help? For the run-or-pass call, we can measure it. Charting services record, for every NFL snap, much of what steps 1 to 5 ask about: the situation, the personnel, the quarterback's alignment and the number of backs, whether there was motion, whether the offense huddled, and how many defenders were in the box. (Steps 6 to 8 aren't in public data: nobody charts pre-snap safety depth or corner leverage for every snap.) So we can do what a reader would do: learn from one season what each combination of clues usually meant, then call every snap of the next season, adding one step at a time.

⁷NFL.com, "The coverage that turned the Seahawks into Super Bowl winners and is taking the NFL by storm" (Next Gen Stats): <https://www.nfl.com/news/the-coverage-that-turned-the-seahawks-into-super-bowl-winners-and-is-taking-the-nfl-by-storm> (two-high on 32.9% of dropbacks in 2018, 42.0% in 2025).

⁸Phil Perry, "Super Bowl 60 preview: Four matchups that will decide Patriots-Seahawks," NBC Sports Boston (Feb. 8, 2026): <https://www.nbcsportsboston.com/nfl/new-england-patriots/super-bowl-2026-preview-matchups-drake-maye-seahawks/766995> ("if we have the ball for 70 snaps in the game, 65 of them are gonna look the same before the ball's snapped").

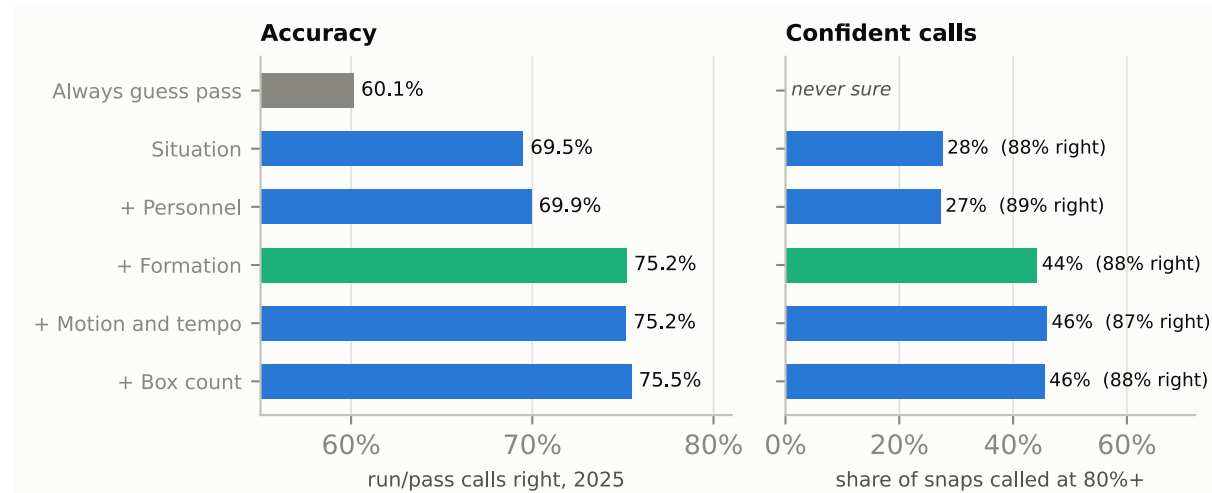


Figure 3: What the measurable steps are worth for a run-or-pass call. Each row adds one checklist step to a model learned on the 2024 regular season and then scored on every 2025 regular-season run and dropback (32,813 snaps). Left: how often the call was right. Right: the share of snaps the model could call at 80% confidence or more, and (in the label) how often those confident calls were right. The situation does most of the work; the formation (quarterback’s alignment and number of backs) is the one big jump after it; motion, tempo and the box add almost nothing to run or pass. Source: nflverse play-by-play, FTN charting and participation.

Three lessons hide in that chart.⁹

The situation is most of the call. Down, distance, field zone, score and clock take you from 60% (“always pass”) to 69%. That is the step a casual fan skips, because it feels too obvious to count. It isn’t obvious; it’s just quick.

The formation is the one big jump after it. Adding the quarterback’s alignment and the number of backs lifts the model to 75%, and, as important, raises the share of snaps it can call at 80% confidence or better from 28% to 44%. Personnel alone added little on top of the situation (about 0.5 points), because most of what the personnel tells you, the formation tells you again, more precisely.

Motion, tempo and the box add almost nothing to run or pass. Together they move the needle by 0.3 points. That isn’t because they’re useless. It’s because they answer *other questions*. Motion is for identifying the coverage. The box is for knowing whether the run, if it comes, has a chance. And the defense’s box count is partly a reply to the offense’s formation, so it repeats what you already knew. The same is surely true of the steps nobody charts: safety depth and corner leverage tell you little about *whether* the offense will throw, and a great deal about *where* and *against what*.

That changes how you should spend your twenty seconds. **By the end of step 3 your run-or-pass call is essentially made. Steps 4 to 8 are for the second layer:** the coverage, the pressure, where the ball should go, which way the run will hit. A reader who spends all twenty seconds agonising over run or pass is using the expensive steps on the cheap question.

The tells catalogue

How to Watch a Broadcast started a catalogue with four starter tells (stance, the back’s depth and offset, splits, safety depth) and promised the full version here. This is it: ten clues, five from each side of the ball, and a plate that shows where each one lives.

⁹Course calculation. Nested logistic regressions (ridge-penalised, fitted with numpy) on nflverse play-by-play, FTN charting and participation. Training set: 2024 regular-season runs and dropbacks; test set: all 32,813 2025 regular-season runs and dropbacks. Situation = down by distance (1, 2, 3, 4–6, 7–10, 11+ yards), field zone, score margin (five bands), last two minutes of a half, and fourth quarter by score; personnel = offensive personnel group from participation; formation = FTN qb_location by number of backs (n_offense_backfield, 0, 1, 2+); motion and tempo = FTN is_motion and is_no_huddle; box = participation defenders_in_box (5 or fewer, 6, 7, 8+). nflverse’s own xpass model, which uses only the situation, calls 71% of the same snaps correctly.



Figure 4: The tells catalogue on one plate. A shotgun offense with 11 personnel (one back, one tight end) faces a nickel defense with two deep safeties. Numbered badges mark where each tell lives: 1 to 5 on the offense, 6 to 10 on the defense; the key below gives the usual meaning in a few words, and the table after the next figure gives the reasons. Every one is a lean, not a guarantee, and some alignments are exaggerated so the clue is visible. Badge 1 sits behind the right tackle, whose stance you read from the side (next figure); Z (4) has cut his split from the numbers (faint circle) to about nine yards from the ball; the left end (6) is aligned well outside the tackle; the right corner is pressed with inside leverage and the left corner is off with outside leverage (9); the Mike has crept to three yards (7); and the right safety (10) is shallower than his partner and pinched to the hash, while the left one stands at normal width. The depth scale on the left counts yards from the line of scrimmage.

Reading the plate offense first, then defense, is deliberate: it is the order of the checklist. Offensive tells belong to step 3, when everyone is set; defensive tells to steps 6 to 8, when the defense has settled (or pretended to). Two of the stance tells are easier to understand from the side:

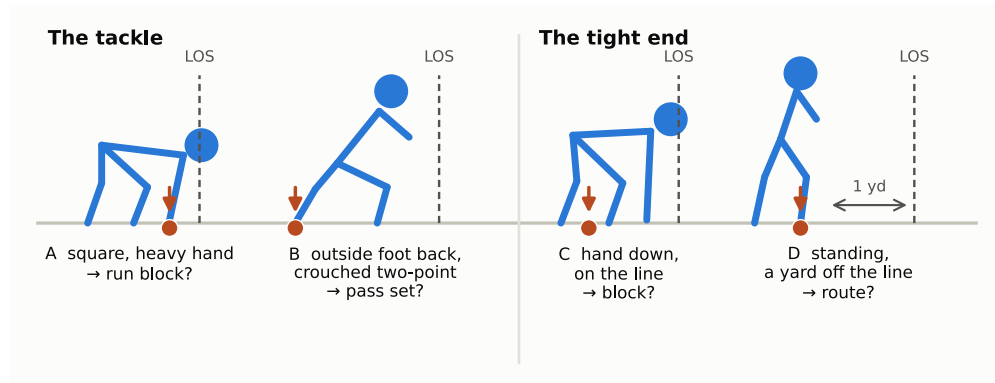


Figure 5: Two stance tells seen from the side. Each player faces right, toward his own line of scrimmage (the dashed line in front of him); the dot and arrow mark where his weight is. A and B, the tackle: about to drive forward, he stands square with his shoulders over a heavy hand; about to kick back into a pass set, he staggers his outside foot well back and sits on it, and on obvious passing downs many tackles go further and stand up in a crouched two-point stance, as here, which is the easiest version to spot on TV. C and D, the tight end: in a three-point stance on the line he is set up to block (though a good one releases from the same stance); standing a yard off the line, he is set up to get into a route quickly. The tight end tell is the weaker one: many teams stand the tight end up on every snap.

Here is the whole catalogue as a reference table: what each tell usually means, *why* it leaks (a tell that has no physical or tactical reason behind it won't survive a self-scout), how much to trust it, and whether you'll see it live or only on the replay.

#	Tell	Usually means	Why it leaks	How far to trust it; where you'll see it
1	Linemen's stance and weight	Heavy hand, weight forward: run. Light hand, hips back, a tackle's outside foot staggered or a two-point stance, the line set a little off the ball: pass. A guard with less weight on his hand, cheated back a few inches or with his heel turned: a pull, so a gap run	A 320-pound man can't explode forward from his heels or kick back from his fingertips, and a puller needs a head start sideways	Medium on early downs; quiet on short yardage, when everyone is heavy. <i>Replay (end zone); live only on close-ups.</i>
2	Back's depth and offset	Shotgun runs usually cross the quarterback's face, away from the back's side; a deeper back wants time to read	The handoff path points one way; reading blocks needs a running start	Medium for direction; weak for run or pass. <i>Live, every snap.</i>
3	Shotgun, pistol or under center	Shotgun mostly pass (79% in 2025); pistol and under center mostly run	Each alignment is built for a family of plays	Strong for run/pass, but under-center passes are mostly play-action, built to exploit this tell. <i>Live, every snap.</i>
4	Receiver split	Reduced: a crack block, a short crosser, or room to break outside. Wide: an isolated route, deep or breaking inside	Receivers set their split to make their job easier	Weak: tells you <i>where</i> to look more than <i>what</i> . <i>Live on the wide shot; replay otherwise.</i>
5	Tight end's stance	Hand down on the line: block likely. Standing, flexed: route likely	Blocking starts low; releasing starts tall	Weak; many teams stand the tight end up on every snap. <i>Live.</i>
6	Defensive lineman's width and stance	Wide-aligned edge: pass rush. A lineman sitting back: stunt or drop	A wide rusher has a straight path to the quarterback; the loopster on a stunt (the lineman who swings around his team-	Medium on obvious passing downs. <i>Replay; live from the end-zone shot.</i>

#	Tell	Usually means	Why it leaks	How far to trust it; where you'll see it
			mate) needs his weight back to go sideways	
7	Linebacker depth	Creeping to 3 yards late: blitz or a downhill run fit. Backing out early: zone drop	A blitzer needs a head start; a dropper needs depth	Medium; walking up is also a bluff. <i>Live, on the wide shot.</i>
8	Nickel's spot	Apexed: run support or zone. Tight on the slot: man, or a nickel blitz	Man coverage needs proximity; an apex keeps both jobs in reach	Medium. <i>Live.</i>
9	Corner depth and leverage	Press: man or a hard corner. Off with outside leverage: help inside (a deep middle safety). Inside leverage: usually man, with no help inside (the sideline is his help) or a safety over the top	A defender takes away the side where he has no help	Medium; leverage also follows the play-call's coverage of the week. <i>Replay (skycam); live on the wide shot.</i>
10	Safety depth and width	Deep and at normal width, 8 to 10 yards from the middle: a deep half or quarter. The shallower one, the one pinched to the hash, or the one moving: the rotation	A safety can't cover more than about eight to ten yards of ground between the snap and the throw	Strong <i>when it moves late</i> ; disguise is the norm. <i>Top edge of the TV picture; skycam replay.</i>

A few of these deserve a sentence more than the table can give them.

The back's offset is a direction tell, not a run-or-pass tell. Shotgun runs most often send the back across the quarterback's face, because the mesh (the moment the ball is handed off) happens while the back is moving across. A back offset to the right usually runs left. But the same offset is where a back sets up to pass-protect against a rusher from that side, so it tells you *which way* a run goes far more than *whether* there will be one. (No public charting records the back's offset, so this one you'll have to calibrate in your own log.)

The run's scheme has one pre-snap tell, and it's on a guard. A gap scheme needs a puller, and a guard who is about to pull sideways often sets up for it: less weight on his hand, a few inches deeper than his neighbours, his heel turned toward where he's going. Zone has no such tell before the snap; you confirm it *at* the snap, when the whole line steps the same way together (Zone Running).

The split tell has two readings, and the book has given you both. A reduced split brings a receiver closer to the formation, which helps three different jobs: a crack block on a linebacker or safety inside him, a short crossing route (the trip across is shorter), and a route that breaks *outside* and needs room to get there. That's why reduced split is a weak tell for *what* is coming. Its real value is a *where*: when a receiver tightens his split on a run look, watch the defender inside him, because that's the man he's coming to block.

Leverage points at the help. A defender usually lines up to take away the side where he has no help, and funnels the receiver toward the side where he has some. So a corner with outside leverage and a single safety in the middle is telling you the safety is his help. Inside leverage is the man-coverage look, and it comes in two kinds. With nobody in the middle (Cover 0, much press Cover 1), the corner takes away the quick inside throw himself and uses the sideline as his help. With two safeties deep, he can sit inside and trail the receiver (trail technique) because a safety is over the top: that's 2-Man. Read leverage as an arrow pointing at the help, and the coverage family usually follows.

Safety width is the most underused tell. Everyone counts deep safeties; few look at how far apart they are. NFL hash marks are only about three yards either side of the middle of the field, so use the numbers as your ruler. A two-high safety at normal width, eight to ten yards from the middle (about halfway between the hash and the top of the numbers), has a deep half or deep quarter to cover and can't reach the deep middle in time. A safety pinched to the hash or inside it, within three to five yards of the middle, *can*, and the one standing shallower than his partner can come down. That asymmetry, not the two-high picture itself, is how you see a rotation coming.

Common misconception

“NFL players are too good to have tells.” Every player has tendencies, because the best way to do a job quickly is to start in the position the job needs. What separates good teams from bad ones isn’t the absence of tells, it’s what they do about them. Every staff self-scouts its own film looking for the clue an opponent could read, and then decides whether to break it, hide it, or keep it as bait (Constraint Theory and Sequencing). The Rams’ under-center passing in 2025 is the textbook bait: they lined up under center on 59% of their snaps (the league: 34%), threw on 40% of those (the league: 31%), and made 81% of those throws play-action. The alignment says run, so the linebackers step up, and the pass goes behind them. A tell that is right 70% of the time is still worth using. You just shouldn’t be 95% sure of it.

Stated confidence: putting a number on it

Step 9 asks for a number, and this is where the course’s most important habit gets its name. A **stated confidence** is the probability you attach to a prediction before the snap, said or written as a number rather than a feeling: not “probably a pass” but “pass, 75%.” The prediction log has had a confidence column since Part 1. From here on, it is not optional.

Why a number, rather than just the call?

1. **It separates a read from a guess, over many calls.** A single miss tells you almost nothing: one 90% call in ten *should* miss. A pile of them is different. If your 90% calls miss one time in three, you’re misreading something; if your 55% calls miss half the time, you were honestly unsure, and you said so. Without the number, your log can’t tell those two patterns apart.
2. **It forces you to use the base rate.** You can’t say “80%” for a pass on first-and-10 without asking what moved you from 50. If nothing did, the honest number is 50 to 55.
3. **It can be scored.** A forecaster who says “70% chance of rain” isn’t wrong when it stays dry; she’s wrong if, across all her 70% days, it rains on nine in ten, or on three in ten. Your calls work the same way: a 70% call should come true about seven times in ten. The next chapter, Film Study, scores the season’s log that way, with a Brier score (the average squared gap between your stated confidence and what happened) and a calibration plot (do your 70% calls come true about 70% of the time?). Researchers who study forecasters have found that saying the number, rather than a word like “probably”, is itself one of the habits that makes forecasters better.¹⁰

How do you choose the number? Start from step 1’s base rate, move it with the strong clues (the formation is the big one), and then respect two limits:

- **Never below 50.** Your call is whichever side you think is more likely, so its confidence is at least 50%. A 50% call is perfectly honest: it says “I have nothing”, and logging it is better than skipping the snap.
- **Rarely above 90.** Offenses choose when to break their tendencies, and the snaps that break them are the ones built to fool you. Even the most lopsided situations leave room for a surprise: third-and-7 to 10 is a pass 94% of the time, not 100%.

What does a well-judged spread of confidences look like? Here is the model from the previous section (steps 1 to 3 only: situation, personnel and formation) on every 2025 snap, grouped by how sure it was:

¹⁰Philip E. Tetlock and Dan Gardner, *Superforecasting: The Art and Science of Prediction* (Crown, 2015), on the forecasting tournaments in which forecasters who used fine-grained numerical probabilities, kept score and updated often were more accurate.

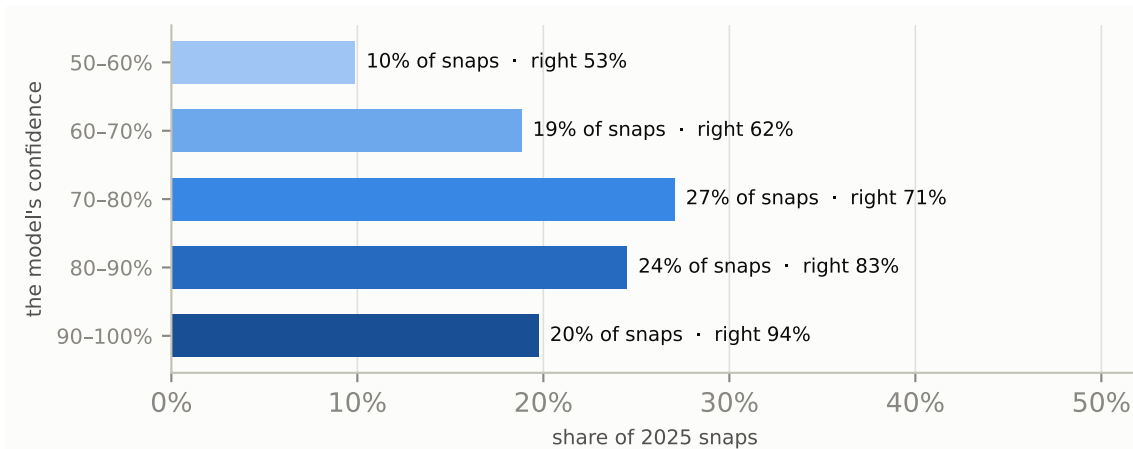


Figure 6: What honest confidence looks like for run or pass. Every 2025 regular-season run and dropback, grouped by how confident a model using only steps 1 to 3 (situation, personnel, formation; learned on 2024) was in its call. Bars: the share of snaps in each band. Labels: how often the call in that band was right. The calls come true at close to the rate they claim (two to four points less often, because the model learned on a different season), and about one snap in five really is a 90%-plus call (long-yardage downs, empty backfields, the end of halves). The rest are spread across the middle, where most of football lives. Source: nflverse play-by-play, FTN charting and participation.

That chart is the target for your own log. When the model says 70 to 80%, it's right 71% of the time; when it says 90% or more, it's right 94% of the time. Nobody expects a person to match a model learned on 33,000 snaps, but the *shape* is the goal: confident where the situation is lopsided, modest everywhere else, and right about as often as you claimed.¹¹

But look at the top band: on a fifth of snaps the model was right 94% of the time. Doesn't a cap at 90 make you *underconfident* there? A little, and that's the point: the two mistakes don't cost the same. On a call that comes true 94% of the time, the Brier score the next chapter uses (lower is better) is 0.056 per call if you say exactly that, 0.057 if you say 90%, and 0.058 if you say 99%. Saying 90 gives up almost nothing; saying 99 costs more even when the situation really is that lopsided, and far more when it isn't, which is the usual human error: in study after study, people's "90% sure" answers come true well under 90% of the time.¹² So 90 is a ceiling for beginners, not a law. When Film Study shows that your 90% calls come true more than nine times in ten, raise your ceiling to 95 for the empty-backfield and third-and-long snaps that earned it.

A small vocabulary helps you say the number fast:

Say	Means	Typical use
"Coin flip"	50%	Nothing to go on: the formation was off screen, or the clues pull both ways
"Lean"	55-60%	First-and-10 from a balanced look; one weak clue
"Probably"	65-70%	The formation agrees with the situation
"Likely"	75-80%	A strong formation clue and a situation that points the same way
"Near certain"	85-90%	Third-and-long, an empty backfield, a two-minute drill when trailing

Layered predictions

Run or pass is the first layer, and after step 3 it's mostly decided. A good reader makes a second call on the snaps where there's time, and states a confidence for it too:

- **On a likely pass:** the coverage family (man or zone; one-high or two-high after the snap), the rush (four, or five or more), or where the ball should go (the quick throw, the sticks, a shot).

¹¹Same model and data as the chart of what each step is worth, using steps 1 to 3 only. Confidence is the model's probability for whichever side it called.

¹²Sarah Lichtenstein, Baruch Fischhoff and Lawrence D. Phillips, "Calibration of probabilities: The state of the art to 1980," in Daniel Kahneman, Paul Slovic and Amos Tversky (eds.), *Judgment under Uncertainty: Heuristics and Biases* (Cambridge University Press, 1982), pp. 306-334: the review that established overconfidence as the usual pattern in people's probability judgments. The Brier figures are a course calculation: a call that says c and comes true at rate q scores $q(1-c)^2 + (1-q)c^2$ on average, with q set to the model's top-band accuracy.

- **On a likely run:** the direction (toward the strength, away from a shotgun back's offset, at the light side of the box) or the scheme family (gap, if a guard is cheating back to pull; zone is confirmed only at the snap, when the line steps together). The hash won't call the direction for you: runs from a hash went to the field and to the boundary about equally in 2025.

Second-layer calls are harder, and your confidence should show it. The base rates help here too. In 2025, defenses sent five or more rushers on only 27% of dropbacks (39% on third-and-7 to 10), so "four rush" is the default until a pressure tell moves you off it. Zone was the coverage on about 69% of charted 2025 dropbacks, so "zone" is the default coverage call, except near the goal line: man rose from 28% of dropbacks outside the red zone to 46% inside it, because a short field leaves little deep space for zone defenders to sink into, and tight coverage on every receiver is the natural answer to a short throw; FTN's man-coverage share has jumped around from season to season in a way that looks like a change in charting rather than in football, so treat that number as rough.¹³ A second-layer call at 60–70% is a good call. Few are better than that, and anyone who claims to call coverages at 90% before the snap is either watching a very predictable defense or not keeping score.

Madden vs. real life

In Madden you pick a play from a menu, and when you're on defense you see the offense's formation and pick a coverage, all with perfect knowledge of your own call. Watching a real game is the reverse: you know neither call, and neither does anyone on the field. The quarterback's pre-snap read is this checklist, run by the person with the most at stake, and he gets it wrong often enough that offenses build hot routes and post-snap reads for the times he does. So a probability isn't a hedge. It is the most accurate description of what anyone, including the players, knows before the snap.

Common misconception

"My 90% call missed, so I was wrong." Not necessarily: one in ten 90% calls *should* miss. You can't judge a single probabilistic call by its outcome, only a pile of them. That's why the log matters, and why Film Study scores a season of calls rather than a play. The mistakes to look for are patterns: 90% calls that miss one time in three (overconfidence), or a log full of 55% calls that come true 75% of the time (you know more than you're admitting).

¹³Course calculation from nflverse participation (defense_man_zone_type) and FTN charting (n_pass_rushers), 2025 regular-season dropbacks with a rush count recorded; coverage share among those charted man or zone. On the season-to-season jumps in FTN's man share (28.8% in 2022 under the earlier NGS source, then 41.2%, 48.3% and 30.8% for 2023–2025), see the course's notes on the 2022/2023 source break in Tendencies.

¹⁴Score, date and site from nflverse schedules and play-by-play (game 2025_21_LA_SEA: Seattle 31, Los Angeles Rams 27, January 25, 2026, Lumen Field); also "2025–26 NFL playoffs," Wikipedia: https://en.wikipedia.org/wiki/2025%E2%80%9326_NFL_playoffs. Seattle's 14–3 record and NFC top seed, and Super Bowl LX (Seattle 29, New England 13, February 8, 2026): https://en.wikipedia.org/wiki/Super_Bowl_LX.

¹⁵Seattle's 2025 staff as listed for Super Bowl LX: https://en.wikipedia.org/wiki/Super_Bowl_LX. Darnold and Kupp (released by the Rams) both signed with Seattle in March 2025: <https://www.si.com/nfl/cooper-kupp-signs-seahawks-rams-release>.

¹⁶nflverse play-by-play, participation and FTN charting for game 2025_21_LA_SEA, Seattle's drive beginning at 9:41 of the third quarter. The charting records personnel (including who was on the field), the quarterback's alignment, backs, motion, tempo, the hash, box count, rushers, coverage label and

Film room: one drive, nine snaps

i Film room: Seahawks 2025 — the NFC Championship Game

The game: Rams at Seahawks, NFC Championship Game, January 25, 2026, at Lumen Field. Seattle, the NFC's top seed (14–3), won 31–27 and two weeks later beat New England 29–13 in Super Bowl LX.¹⁴ In 2025 Seattle's head coach was Mike Macdonald, its offensive coordinator Klint Kubiak (whose system comes from the Shanahan family tree; see Shanahan's Wide Zone Tree) and its quarterback Sam Darnold.¹⁵

The drive: Seattle had gone up 24–13 early in the third quarter; the Rams answered with a touchdown to make it 24–20 with 9:45 left in the quarter. After the touchback, Seattle drove 65 yards in nine snaps and scored on third-and-3 from the Rams' 13, a 13-yard pass to Cooper Kupp, to go up 31–20 with 4:52 left in the third.

Why this drive: it has a bit of everything the checklist is for: three different personnel groups, under center and shotgun, an empty backfield, a screen against a blitz, play-action, a no-huddle snap, two third downs and a red-zone touchdown, against a defense that played base, nickel and dime and three different coverage labels.

The sources: every fact below comes from the public charting in nflverse;¹⁶ the diagrams are typical pictures of what was charted, not the film.

Here is the drive as your log would record it. The calls weren't made by watching the game. They were made by rule from the pre-snap columns only, so hindsight can't creep in: the run-or-pass call and its confidence come from the step 1 to 5 model, fitted on the 2024 and 2025 regular seasons; the coverage and rush calls are the 2025 base rates for that down, distance, field zone and number of defensive backs.

PREDICTION LOG • NFC Championship, Rams at Seahawks • Seattle offense, Q3								
#	Situation	Offense shows	Defense	Call	If pass	What happened	R/P	Pass layer
1	1st & 10 SEA 35 • L hash	12 • UC • 1 back motion	base box 7	Run 55%	zone 80% 4 rush 65%	Holani run, +3	✓	–
2	2nd & 7 SEA 38 • R hash	12 • UC • 2 backs motion	nickel box 6	Run 60%	zone 75% 4 rush 80%	Walker run, –2	✓	–
3	3rd & 9 SEA 36 • R hash	11 • gun • 1 back hash	dime box 5	Pass 90%	zone 65% 4 rush 75%	Kupp catch, +12 (1st)	✓	✓ ✓
4	1st & 10 SEA 48 • L hash	13 • UC • 2 backs motion	base box 7	Run 70%	zone 80% 4 rush 65%	Walker run, +11 (1st)	✓	–
5	1st & 10 LA 41 • middle	11 • gun • empty	nickel box 6	Pass 85%	zone 80% 4 rush 80%	Holani catch, +9	✓	✓ ✓
6	2nd & 1 LA 32 • R hash	11 • gun • 1 back motion	nickel box 6	Run 55%	zone 70% 4 rush 75%	Smith-Njigba screen, +12 (1st)	✗	✓ ✗
7	1st & 10 LA 20 • L hash	12 • UC • 2 backs motion	base box 7	Run 75%	zone 65% 4 rush 55%	Play-action, Barner +7	✗	✓ ✓
8	2nd & 3 LA 13 • L hash	12 • UC • 1 back motion, no-huddle	base box 7	Run 70%	man 70% 5+ rush 80%	Incomplete to Bobo	✗	✗ ✓
9	3rd & 3 LA 13 • L hash	11 • gun • 1 back motion	nickel box 6	Pass 75%	man 55% 4 rush 75%	Kupp, 13-yard TD	✓	✗ ✓
Run/pass: 6 of 9 right. Pass layer on the 6 dropbacks: coverage 4 of 6, rushers 5 of 6.								

Figure 7: The drive as a prediction log: Seattle's nine snaps from its own 35 to the end zone, NFC Championship Game, third quarter. Situation, offense and defense are the pre-snap facts in the charting (L/R hash = the ball spotted on the offense's left or right hash, so the field is the other way; UC = under center; M = motion; NH = no-huddle; box = defenders in the box). The calls were made by rule from those facts alone: run or pass from the checklist model, the 'if pass' layer (coverage family, rushers) from 2025 base rates for the situation and the number of defensive backs. The 'if pass' layer is filled in on every snap, because you make it before you know whether the ball will be thrown; the last two columns score the calls, and the pass layer is scored only on dropbacks, because coverage isn't charted on runs. Source: nflverse play-by-play, participation and FTN charting, game 2025_21_LA_SEA.

time to throw, but not exact alignments or the routes of receivers who weren't thrown to. Charting is done by people and contains errors; the labels are FTN's, not the teams'.

And here is how each call was built, step by step, the way the confidence should move in your head:

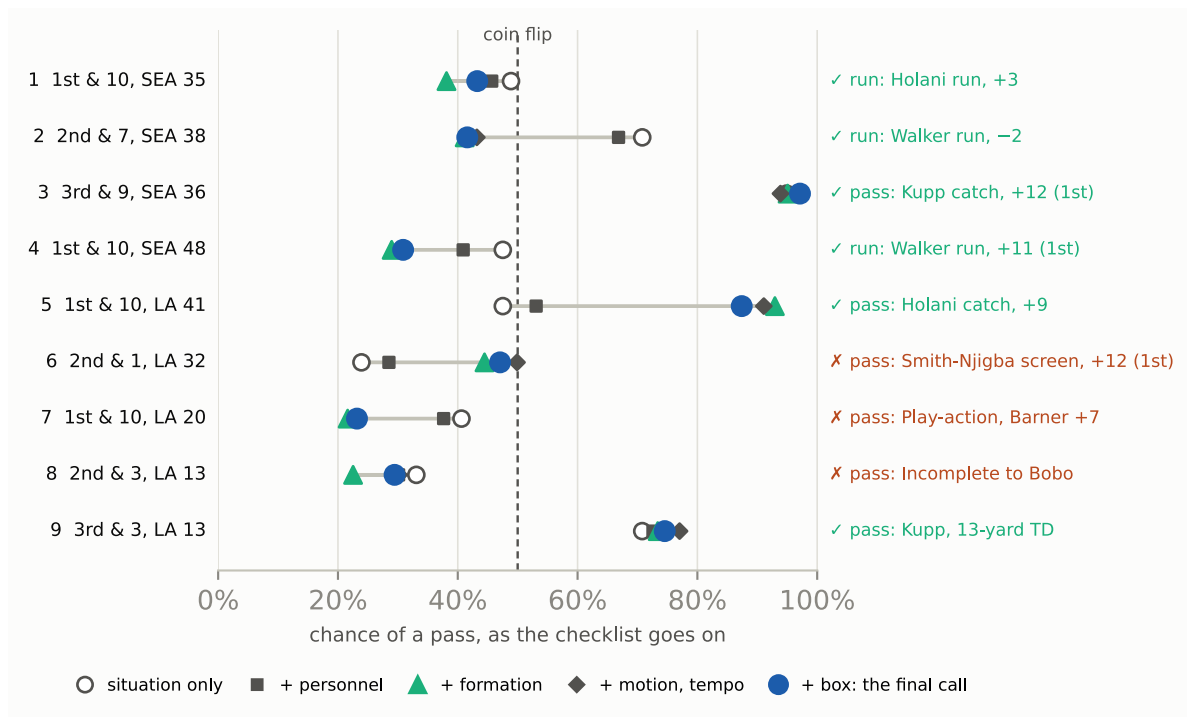


Figure 8: How the run-or-pass confidence moved through the checklist on each snap of the drive. Each row is one snap, and each marker adds one step: the hollow circle is the probability of a pass from the situation alone, the small square adds personnel, the triangle adds the formation (quarterback alignment and number of backs), the diamond adds motion and tempo, and the large filled circle adds the box count and is the final pre-snap call. The right-hand label is what happened, ticked or crossed against the call. The biggest moves come at the triangle, the formation step: snap 5 (an empty backfield) jumps toward pass and snaps 2, 4 and 7 (two backs under center) drop toward run. The misses are the second-and-1 screen (a coin flip that landed the other way) and snaps 7 and 8, passes from under center with two tight ends, exactly the look offenses use to break the tendency. Model fitted on the 2024 and 2025 regular seasons; source: nflverse play-by-play, participation and FTN charting.

And the drive in pictures, one pre-snap look per snap:

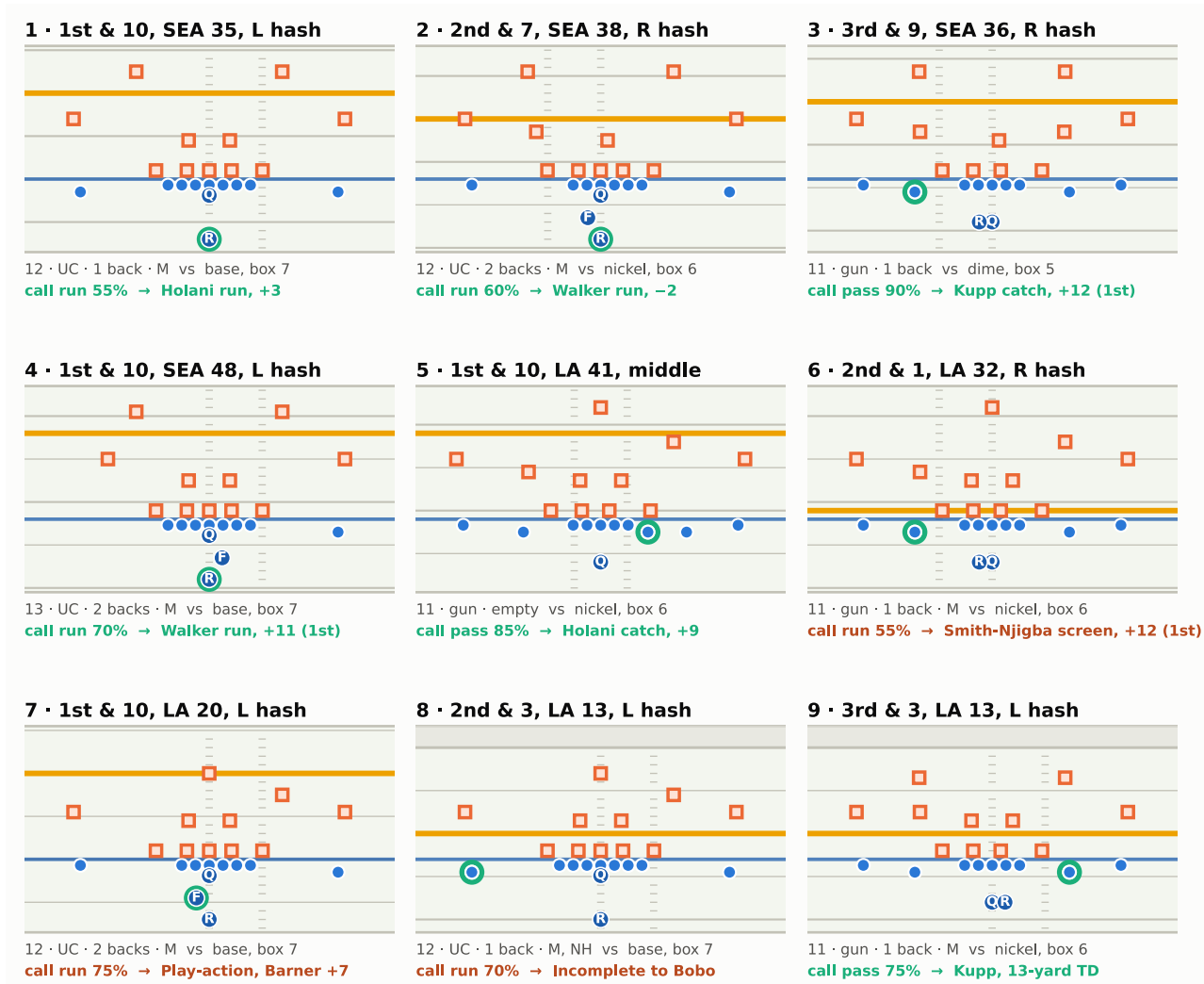


Figure 9: The drive in pictures: a typical pre-snap look for each of the nine snaps, built from what was charted (offensive personnel, quarterback alignment, number of backs, defensive personnel, box count and the hash, all exact), with the ball on the right yard line and hash and the yellow line to gain. The players in the backfield are drawn dark and lettered (Q quarterback, R running back, F a fullback or a tight end lined up as an H-back), because one back or two is the formation clue on most of these snaps. Everything else is illustrative: the charting doesn't record exact alignments, and for runs the safeties are drawn two-high because the coverage isn't charted on runs; for passes the shell matches the charted coverage family. In the red zone (snaps 7 to 9) the secondary is drawn closer to the line, because the end zone is behind it. The ringed player is the ball carrier or the receiver the ball went to. Offense: blue circles; defense: orange squares.

Now walk it the way you would live, step by step. The snap numbers match the log.

Snap 1: first-and-10 at the Seattle 35, left hash. A four-point lead in the third quarter, a coin-flip down. Seattle sends in 12 personnel (two tight ends; Kupp and Jake Bobo are the receivers) and the Rams stay in base. Under center, one back, motion: a run look, but the call is only run, 55%, because offenses throw from this look too. Seven in the box against seven blockers is a fair fight. George Holani runs behind the right guard for three. ✓

Snap 2: second-and-7. Second-and-medium leans pass, but Seattle stays in 12 personnel with a second player in the backfield, most likely a tight end lined up as an H-back (in the backfield, just off the tackle's hip). Two backs under center swings the call back to run, 60%. The Rams answer with a nickel, which looks like the defense betting on a pass, until you notice *who* came on. Snap 1's receivers were Kupp and Bobo; snap 2's are Kupp and Jaxon Smith-Njigba, the league's 2025 Offensive Player of the Year. On this drive every snap with Smith-Njigba on the field drew a nickel or a dime, and every 12-personnel snap without him drew

¹⁷Personnel and the players on the field from nflverse participation (`offense_players`, `defense_personnel`) for the drive's nine snaps, with names from nflverse rosters. Smith-Njigba's award: NFL.com, "List of NFL Honors award winners from 2025 NFL season": <https://amp.nfl.com/news/list-of-nfl-honors-award-winners-from-2025-nfl-season>.

base.¹⁷ The Rams were matching a player, not a digit: that's step 2's "notice *which* receivers". Kenneth Walker loses two around the right end. The call was right; the run wasn't, and you're scoring your read, not the offense.

Snap 3: third-and-9 at the 36, right hash. The run-or-pass question answers itself (pass, 90%), and everything after step 3 goes into the second layer. Seattle goes to 11 personnel in the shotgun; the Rams go to dime (six defensive backs) with only five men in the box, a defense that wants to keep everything in front of it and tackle short of the sticks. The base-rate calls say zone and a four-man rush, and that's what came: four rushers, and a zone FTN labelled Cover 9, a number that means different things to different staffs (Coverage Fundamentals); the figure draws the Fangio meaning, a weak-rotation Cover 3 (Fangio's Two-High Revolution). Darnold held the ball 2.8 seconds and found Kupp short to the left, toward the field, a throw FTN charted as a checkdown: six yards in the air and six after the catch, a first down by three yards.

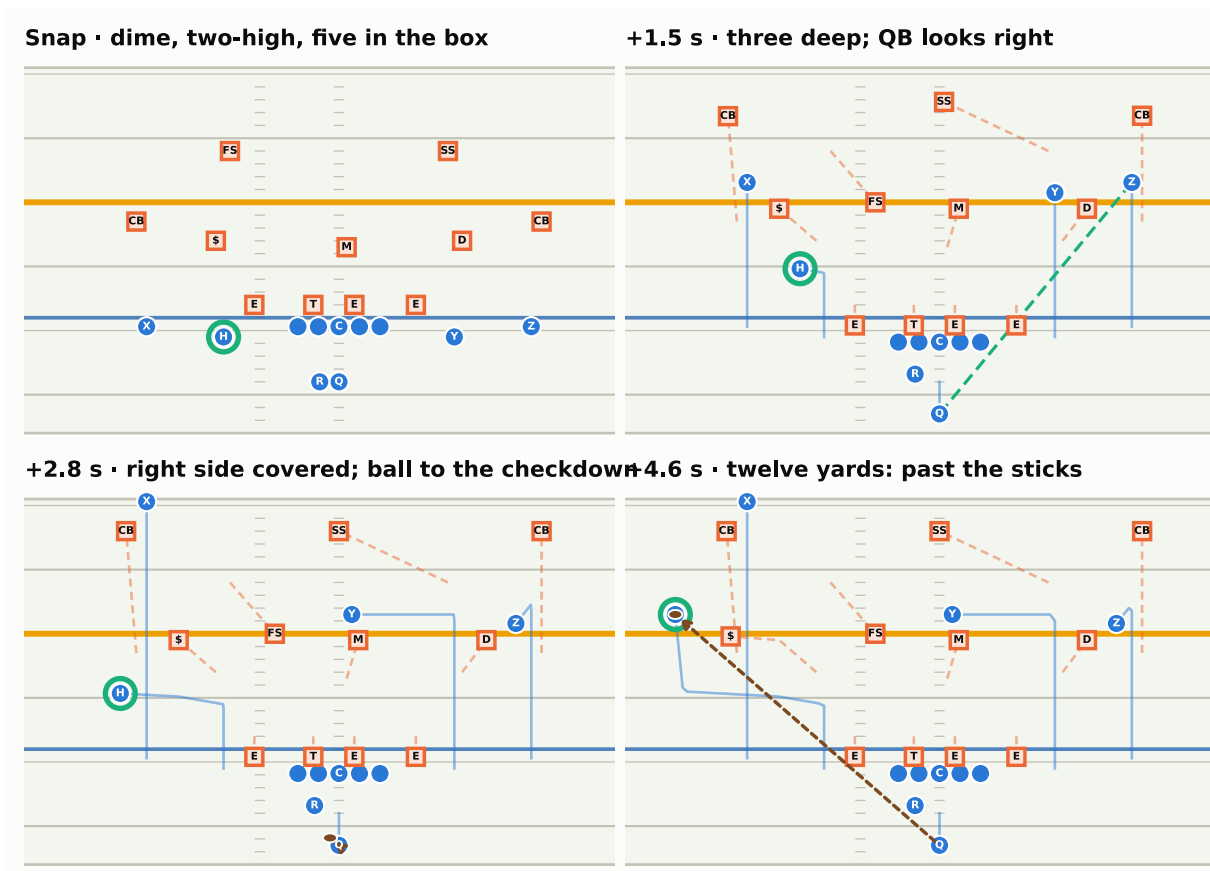


Figure 10: Snap 3, third-and-9 from the right hash (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun, no motion; the Rams in dime with five in the box; four rushers; a zone labelled Cover 9; a checkdown thrown after 2.8 seconds to Cooper Kupp (H, ringed) on a short out to the left, toward the wide side of the field: six yards in the air and six after the catch for the first down. Illustrative: the alignments, the other routes, and the rotation, drawn as the weak-rotation Cover 3 meaning of 'Cover 9' (two-high before the snap, the safety away from the tight end coming down, three deep). Defensive letters: E, an end or edge rusher; T, a tackle; M, the one linebacker; \$ and D, the nickel and dime backs; FS and SS, the safeties. What to notice: the quarterback's eyes (the green dashed line) start on the right, where Z's curl and Y's dig are covered by the underneath defenders sinking to the sticks, and only then does the ball go left; the conversion comes from the yards after the catch.

The lesson of snap 3 is the most common lesson of third-and-long. The run-or-pass call is free; the defense's plan is easy to read from its personnel; and the play is decided by the thing you can't predict, a receiver breaking a tackle or not. Your second-layer call ("zone, four rush, the ball goes short and the game is the tackle") was right, and that's as good as reading gets.

Snap 4: first-and-10 at the 48, left hash. Seattle goes heavy: 13 personnel (three tight ends, one receiver), under center, two backs, motion. The Rams match with base and five linemen and edge rushers up front (two of them listed on the roster as nose tackles), a seven-man box. The call is run, 70%, as confident as a first-down run call gets, and that leaves time for a second layer: *which way*? Under center there's no shotgun offset to read, so the clues are the strength (the side with more tight ends, or the H-back's side) and the light side of the box. The charting doesn't record which side those were, so this drive can't score a direction call. Walker went behind the right guard for eleven. In your own log, write the direction call anyway (12 right, 0.6): your eyes record what the charting doesn't.

Snap 5: first-and-10 at the Rams' 41. The reverse picture: 11 personnel, shotgun, and **nobody in the backfield**. An empty backfield is the strongest pass tell in the catalogue (95% of 2025 empty-backfield snaps were dropbacks), and the call is pass, 85%. The second layer is the interesting part. Empty means five receivers and only five blockers, so if a defense brings a sixth man nobody can block him, and the quarterback has to beat him with a quick throw; that's why empty is so often a quick-throw formation. The Rams rushed four and played Cover 3, and Darnold checked the ball down to Holani, the running back, who had lined up outside the backfield as a receiver, on a shallow crossing route for nine.

Snap 6: second-and-1 at the 32, right hash. The most interesting snap of the drive. Second-and-short is the down where offenses take chances, because even a failed play leaves a manageable third down; from the shotgun with 11 personnel it was a pass 57% of the time in 2025. The model says run, 55%. Why the disagreement? The model adds the down and the formation as separate pushes and never learns that this particular pairing, the shotgun on second-and-short, throws more than the two pushes suggest: at the formation step it says 44% where the raw rate is 57%. Narrow the raw rate to this exact look (one back, motion, six in the box) and it falls to 49%, though on only 57 snaps. Either way it's close to a coin flip, and there's a general lesson: when you know the base rate for the exact look, trust it over a sum of clues. The second layer said zone (70%) and a four-man rush (75%). The Rams played zone but sent five, and Darnold got the ball out in 0.7 seconds on a screen to Smith-Njigba, who caught it four yards *behind* the line, toward the field, and ran sixteen yards after the catch. **X** on run or pass, and a 55% call is allowed to miss: that is what 55% means.

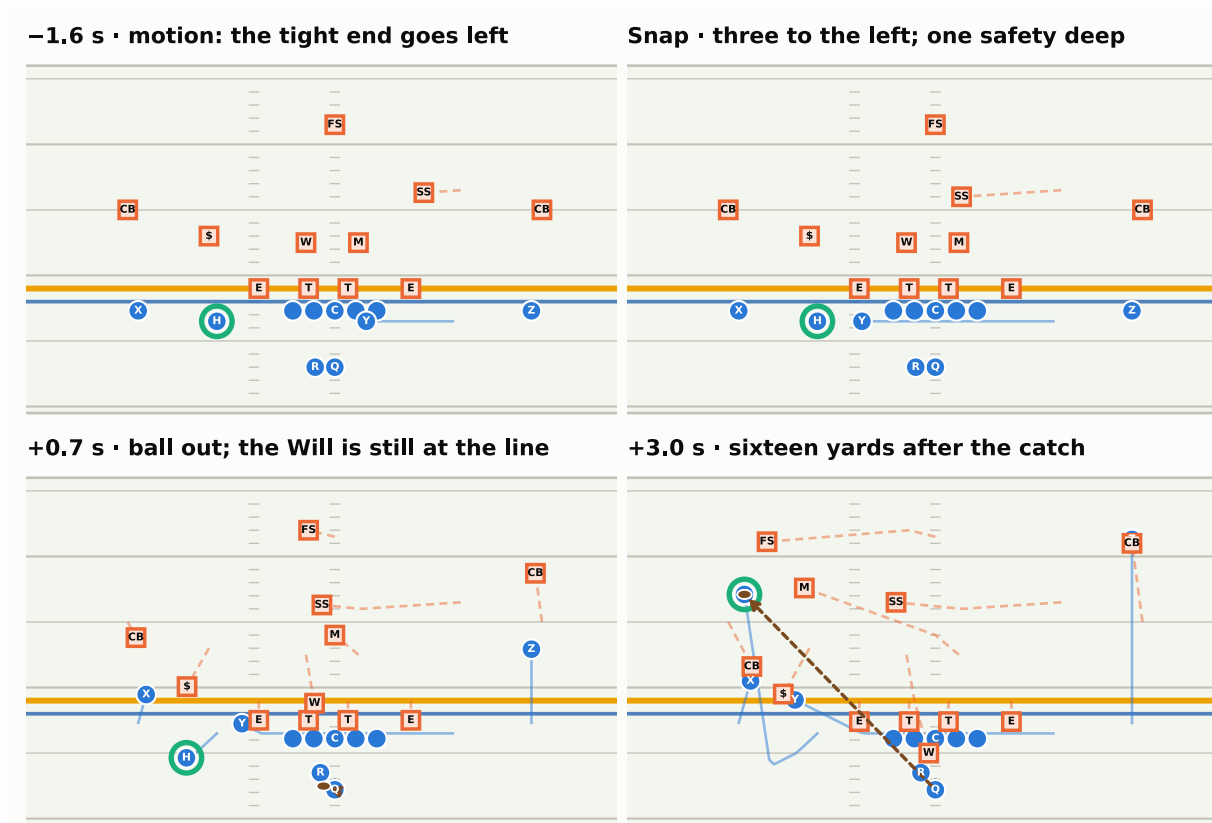


Figure 11: Snap 6, second-and-1 from the right hash (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun with motion; the Rams in nickel with six in the box; five rushers (one of them a blitz) and Cover 3 behind them; a screen to Jaxon Smith-Njigba (H, ringed) on the left, toward the field, thrown 0.7 seconds after the snap, caught four yards behind the line, sixteen yards after the catch, tackled by an inside linebacker. Illustrative: who motioned (here the tight end, Y, across to the left), which defender blitzed (here the Will linebacker, W), and the blocks. What to notice: the third panel, when the ball is already out and the blitz has only reached the line; with him gone, the screen side has three Seattle players (X and Y blocking, H with the ball) against two Rams defenders, the corner and the nickel.

Why does a screen beat a blitz? Because every extra rusher is a defender who isn't covering anyone. This was a perimeter screen, thrown at once: the blitzing Will was one fewer Ram on the outside, so the screen side had three Seattle players against two defenders, and the ball was gone before the fifth rusher mattered. A pressure call against a screen is the worst outcome for a defense, and it happens because the defense was guessing too: second-and-1 from the shotgun is a coin flip for everyone. Could step 8 have warned you? The charting can't say whether the Will showed his blitz before the snap, so the reconstruction doesn't

draw a tell. If he did (a linebacker creeping toward the line with one safety deep), your second-layer call should have moved from “four rush” toward “five or more, ball out fast”.

Snap 7: first-and-10 at the Rams’ 20. Back to 12 personnel, under center, two backs, motion, against base and a seven-man box. Every step says run: run, 75%. It was play-action. Darnold faked the handoff, left the pocket and, after 4.7 seconds, found the tight end AJ Barner short on the left for seven. ✗. This is the miss you *should* have, the tendency-breaker built to beat the reader who believes the look: an offense from the Shanahan tree lives on making its run and play-action looks identical (Play-Action, Bootlegs and Screens). At 75% the call was honest. At 95% it would have been a misread. And the model doesn’t know it’s Seattle; you would. From 12 personnel under center on first down, Seattle threw 42% of the time in 2025 (139 snaps), against 34% for the league, so a reader who copied the team’s tendencies before kickoff would have started this snap several points closer to pass.

Snap 8: second-and-3 at the 13, no huddle. Seattle hurried to the line in the same personnel, under center with one back. The call is run, 70%, and it missed again: a quick throw to Jake Bobo, out 0.6 seconds after the snap and caught 1 yard behind the line, which FTN charts as a designed throw with no progression: a perimeter extension of the run game rather than a dropback. Incomplete. The second layer split: in this spot (second-and-short in the red zone, four defensive backs) the 2025 base rates said man (70%) and five or more rushers (80%). The Rams did send five, but played Cover 3 behind them. Two run-call misses in a row, both passes from a run look, is not bad reading; it’s a play-caller deliberately working against his own tendency in the red zone. The between-snaps habit (line 13 on the card) is where you’d note it: *12 personnel under center, inside the 20, two passes in a row*. If they come back to it, your next call from that look should drop toward 55%.

Snap 9: third-and-3 at the 13, left hash. Shotgun, 11 personnel, motion: the model says pass, 75%. Third-and-short in the red zone is one of the few spots where man coverage is about as common as zone (the 2025 base rate for this spot was man 55%), so the coverage call was close to a coin flip. The Rams played zone, a Cover 6 (quarters to one side, Cover 2 to the other), set the way Cover 6 usually is: quarters to the field, on the right, where Kupp was. They rushed four and got interior pressure, but Darnold threw on time, 2.4 seconds after the snap, to a receiver in his progression rather than a checkdown:¹⁸ Kupp on a short post, caught at the 6, six more yards after the catch, touchdown. ✓

¹⁸FTN’s `read_thrown` field (nflverse FTN data dictionary: https://nflreadr.nflverse.com/articles/dictionary_ftn_charting.html) codes this throw as a read in the progression, not a checkdown or a designed throw; which read is ambiguous in the 2025 file, and Film Study shows how to check calls like this one on the All-22.

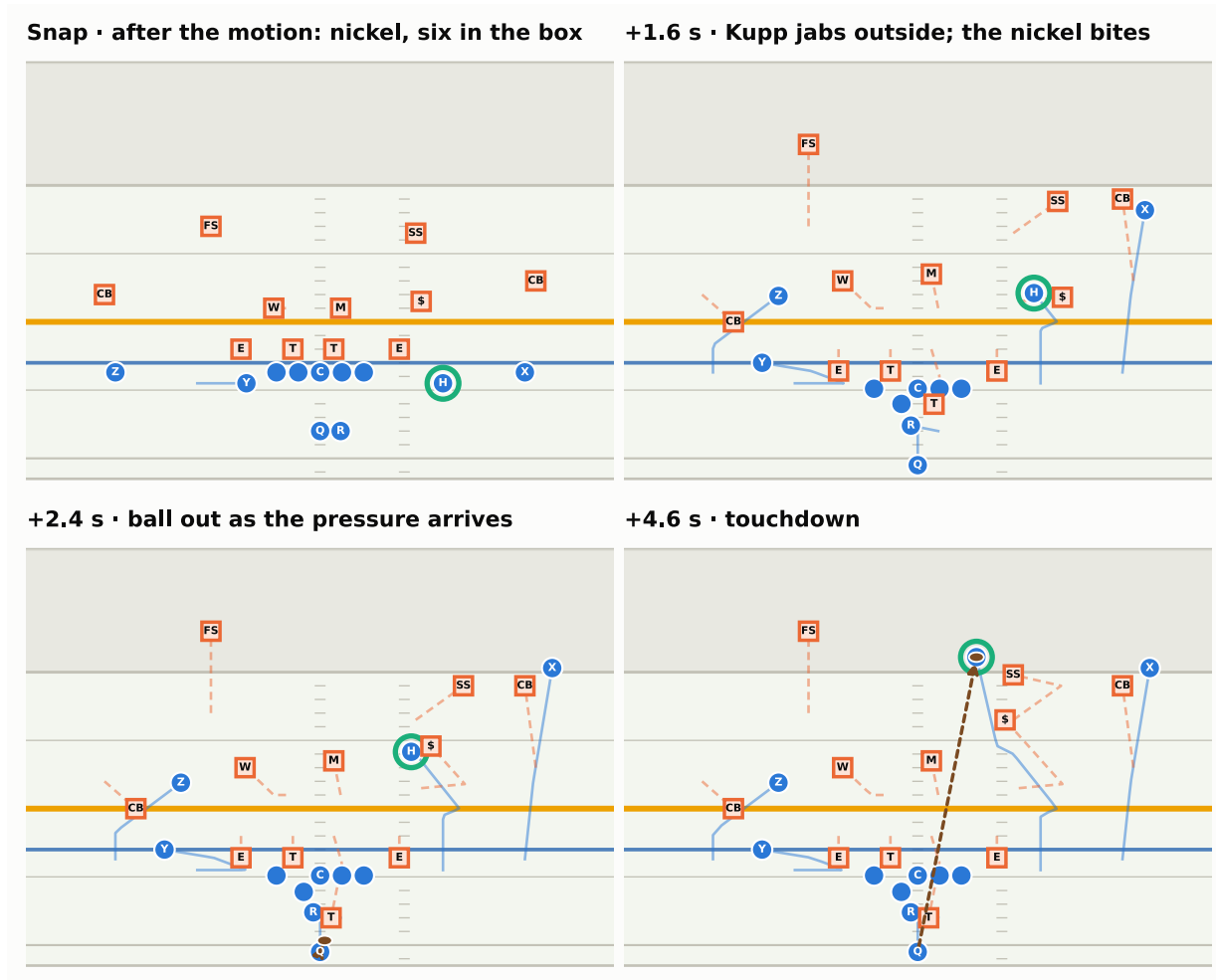


Figure 12: Snap 9, third-and-3 at the Rams' 13, from the left hash (a reconstruction, simulated in Big Data Bowl format). Charted facts: 11 personnel in the shotgun with motion; nickel, six in the box; four rushers, with interior pressure (charted: Kobie Turner, a defensive tackle, hit Darnold); Cover 6; the ball thrown inside the progression (not a checkdown) to Cooper Kupp (H, ringed) on a post from the right side, toward the field, seven yards in the air and six after the catch for the touchdown. Illustrative: the alignments, the tight end's motion (Y), the other routes, the rusher's path, and the coverage drawn as quarters on Kupp's side and Cover 2 on the other. The nickel (\$) bites on Kupp's jab outside, and the quarters safety on that side widens to help on the fade outside him and comes back too late. What to notice: the third panel, the ball leaving as the tackle comes through the B gap; and that in the red zone the deep defenders are already near the goal line, so a short post that gets inside the nickel and in front of the safety has room to run.

Add it up. The run-or-pass calls went 6 for 9. The three misses were all passes from looks that lean run: a coin flip on second-and-1, then play-action and a quick throw from under center with two tight ends, the looks an offense uses to spend its tendency. On the 6 dropbacks the second layer went 4 for 6 on the coverage family and 5 for 6 on the rush count. The coverage misses both came in the red zone, where the base rate leaned man and the Rams played zone; the rush miss was the second-and-1 blitz. That's a realistic live result for a good reader, and it says where to look harder: in the red zone, don't trust the base rate for coverage, because that's where the motion response and the shell earn their seconds.

Three habits from the drive are worth keeping:

- **Score the read, not the play.** Snap 2 was a correct run call on a run that lost yards. Snap 3 was a correct pass call on a checkdown that only converted because of the yards after the catch. Your log grades you on what you could know before the snap.
- **Expect the tendency-breaker where it is cheapest.** Play-action from a heavy look on first down, and a quick pass from the same look in the red zone, are the classic places an offense spends a tendency. A 65% call that misses there is fine; a 90% call that misses there is a lesson.
- **Notice the defense's bets.** Dime with five in the box on third-and-9, a nickel the moment Smith-Njigba came on, five rushers on second-and-1: each is the defense making its own prediction, and each is a clue to the second layer.

i Go deeper: the same drive in real tracking data

The figures above are simulations in the Big Data Bowl's tracking format, built from the charting. The NFL releases real player-tracking data only for the seasons and weeks of each year's Big Data Bowl, and no public release covers the 2025 playoffs (the 2026 edition used the 2023 regular season). For a play that is in a release, the real version is one call to `gridiron.bdb.prepare()` away (the code is in the web version of this page), and Tracking Data and the Big Data Bowl walks through it.

Between snaps: keeping the read up to date

The checklist is per-snap, but the best clue about the next snap is often the last twenty. Two habits keep your read current, and both happen while the broadcast is doing something else.

After every snap (five seconds, during the replay): write the log line. The call, the confidence, the clue that decided it, the result. That's line 12 on the card, and it's the habit How to Watch a Broadcast started. Don't skip the clue column: after a few games it tells you which tells are working *for you*.

At every stoppage (a TV timeout, the end of a quarter, a change of possession): update the picture of this team, today. Three questions:

1. **What are they doing from each look?** "Three runs from 12 personnel under center on first down" is a game tendency. It is more current than any season number, and play-callers know it too, so expect the tendency breaker when the streak gets long. That's the honest version of the "they're due" instinct: not that a pass is owed, but that the play-caller is watching the same tally you are.
2. **What did they just change?** Staffs adjust between series, not just at halftime: a new protection against a rusher who is winning, a coverage rolled toward a receiver who is hurting them, a run that worked coming back with a twist (In-Game Adjustments and Tempo). The call after a successful play is often built on it.
3. **What does the clock want?** The score and the time left change step 1's base rate (a team ahead late runs, a team behind throws). And when fourth down comes, make it its own prediction before the offense decides: go, kick or punt, with a confidence, logged in the second-layer columns (`l2 go, l2conf 0.6`).

When the offense goes fast, or the camera won't show you

Against tempo the twenty seconds shrink to five or ten, and the full checklist won't fit. Run the short one: **1, 3, 6, 9**. Situation (you already have it from the last snap, plus the new down and distance), formation (the single biggest clue after the situation), shell (the single most useful second-layer clue), and the call. Drop personnel (it usually hasn't changed; that's the point of tempo), motion (tempo offenses use less of it), and the box and leverage counts you can't finish anyway. The measured value of what you drop is small for run or pass; the cost is mostly in your second layer, so lower those confidences when the offense is hurrying.

The same applies when the broadcast is still on a replay at the snap, which happens more than it should. Call it from the situation alone, write "sit" as the clue, and move on. A 60% call from the scorebug is still a call.

The commonest problem is milder: the snap is live, but the shot is tight, and the safeties and corners are off the top and sides of the screen. Steps 1 to 5 still work (the formation and the box are in the picture); steps 6 and 7 don't. Write ? in the `shell` column rather than guessing, make your second-layer call from the base rate alone, and keep its confidence at 60% or below. Then use the replay: the end-zone and high angles usually show the whole secondary, so fill in the shell afterwards and note that you saw it late. Coverage calls you couldn't see live get checked after the game, on the All-22, as Film Study shows.

The log, final form

Your log has grown a column at a time since Part 1. Here is its final form, the one Film Study will score. Keep the columns from How to Watch a Broadcast (snap, quarter, clock, down, to-go, ball on, score) and add or tighten these:

Column	What you write	Example
<code>pers</code>	Offensive personnel, two digits	<code>12</code>
<code>align</code>	QB alignment and number of backs	<code>U1</code> (under center, one back), <code>S0</code> (empty)
<code>motion</code>	Motion or shift before the snap	<code>M</code> or blank

Column	What you write	Example
box	Defenders in the box, if you counted	7
shell	One-high or two-high, pre-snap; ? if it was off screen	1, 2 or ?
call	Run or pass	R
conf	Your stated confidence, as a decimal	0.65
clue	The one thing that decided it	align
l2	Your second-layer call (on fourth down: go, kick or punt)	zone, 5+, left, go
l2conf	Its confidence	0.6
result, l2result	What happened, filled in after the snap	P, zone

At halftime, a few lines of pandas tell you how the half went. Here it is run on the drive above, typed into a CSV file called `nfc_title_q3.csv` with these nine lines (copy them if you want to run it yourself):

```

snap,down,togo,pers,align,call,conf,clue,result
1,1,10,12,U1,R,0.55,align,R
2,2,7,12,U2,R,0.6,align,R
3,3,9,11,S1,P,0.9,align,P
4,1,10,13,U2,R,0.7,align,R
5,1,10,11,S0,P,0.85,align,P
6,2,1,11,S1,R,0.55,align,P
7,1,10,12,U2,R,0.75,align,P
8,2,3,12,U1,R,0.7,align,P
9,3,3,11,S1,P,0.75,align,P

```

```

import pandas as pd

log = pd.read_csv("nfc_title_q3.csv")
log["hit"] = log["call"] == log["result"]
print(f"Right on {log.hit.sum()} of {len(log)} calls ({log.hit.mean():.0%}); average stated confidence {log.conf.mean():.0%}")
bands = pd.cut(log.conf, [0.5, 0.6, 0.7, 0.8, 0.91], right=False, labels=["50-59", "60-69", "70-79", "80-90"])
print(log.groupby(bands, observed=True).hit.agg(calls="size", right="mean").round(2).to_string())

```

Right on 6 of 9 calls (67%); average stated confidence 71%

	calls	right
conf		
50-59	2	0.5
60-69	1	1.0
70-79	4	0.5
80-90	2	1.0

Two numbers to watch over a season, not a half: your hit rate against your average stated confidence (if you're right 70% of the time while claiming 80%, you're overconfident), and the hit rate within each band (your 80% calls should come true about 80% of the time). Nine snaps can't tell you either; a few hundred can, and that is what the next chapter does with them.

💡 Watch for it on Sunday: the live run

For **one half** of one game, watching one offense (one team's play-caller, so the between-snaps tally is about one set of habits; switch teams the next time):

- **First, a warm-up.** Before kickoff, pause a recorded game the moment an offense sets, give yourself ten seconds, and say steps 1 to 9 out loud. Do it on five snaps. The order should feel automatic before the clock is real.
- **Before every snap,** run the card in order: situation, personnel, formation, motion response, box, shell, leverage, pressure, call. Say the call and a confidence between 50% and 90% out loud *before* the snap. If the broadcast is still on a replay, call it from the situation and write "sit".
- **On the snaps where you have time,** add one second-layer call with its own confidence: on likely passes, the coverage family or the rush (four, or five or more); on likely runs, the direction.
- **During the replay,** write the line: down, to-go, pers, align, call, conf, clue, and the second-layer call if you made one. Fill in the result.
- **At each stoppage,** update the tally of what this team does from each look today, and note anything it changed.
- **When the offense goes no-huddle,** switch to the short checklist (1, 3, 6, 9) and lower your second-layer confidences.
- **At halftime,** score it: hit rate, average confidence, and the hit rate of your 80%-plus calls. Keep the sheet. Film Study adds a second half of work to it: the All-22, the chart, and the score.

Predict the play

💡 Predict the play 1: first down, two backs

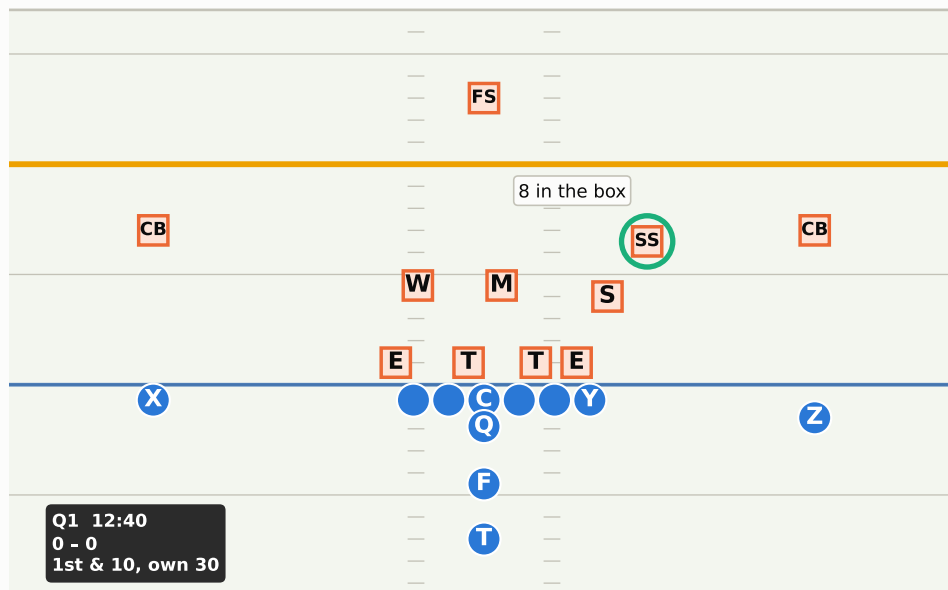


Figure 13: First quarter, 0–0. First-and-10 at the offense's own 30. The offense has 21 personnel (two backs, one tight end) in an I-formation: quarterback under center, fullback (F) and tailback (T) stacked behind him, the tight end (Y) on the right. The defense is in base with the strong safety (ringed) walked down to the edge of the box: eight defenders in the box, one safety deep in the middle.

Run or pass, how sure are you? And a second layer: if it's a run, which way; if it's a pass, what kind?

Answer 1 is at the end of the chapter. Try it first.

💡 Predict the play 2: the mugs on third down

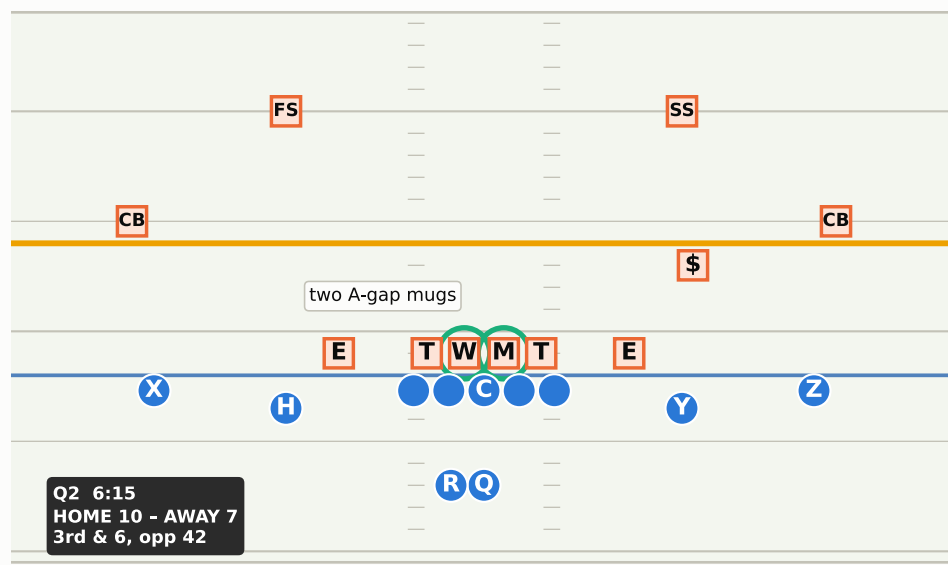


Figure 14: Third-and-6 at the opponent's 42, second quarter. Shotgun, 11 personnel, two receivers each side, the back to the quarterback's left. The defense shows a pressure look: both inside linebackers (ringed) have walked up into the A gaps, either side of the center, so six defenders are at the line. Behind them, two safeties stand 12 yards deep and both corners are seven yards off.

It's a pass; how many will rush, and where should the ball go?

Answer 2 is at the end of the chapter. Try it first.

💡 Predict the play 3: tempo after a first down

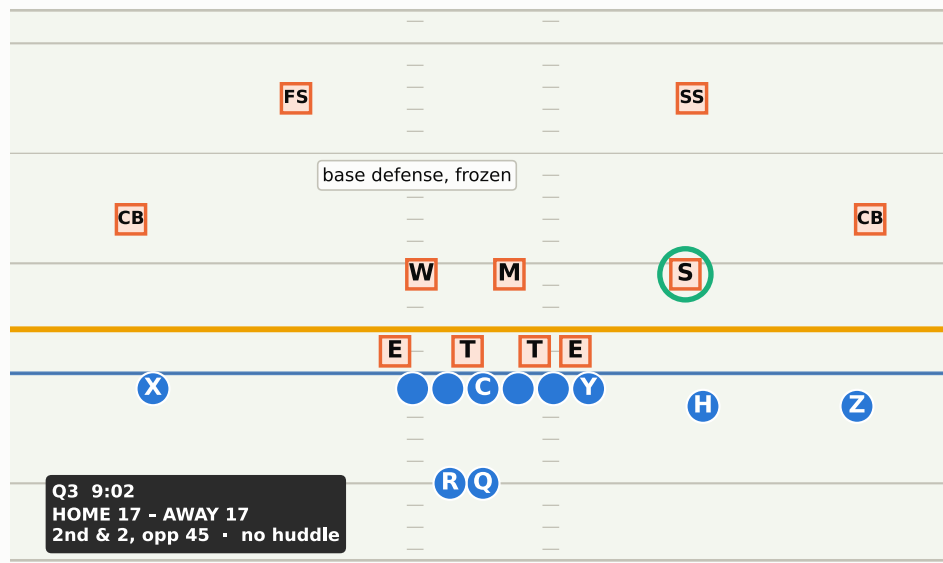


Figure 15: Second-and-2 at the opponent’s 45, third quarter. The offense just ran for 8 yards on first down out of 11 personnel and sprinted to the line without huddling or substituting. Now it shows trips right: the tight end (Y) attached to the right tackle, a slot receiver (H) and Z outside him, with X alone on the left. The defense, which had its base personnel on the field (three linebackers, two corners, two safeties), couldn’t substitute: its strongside linebacker (S, ringed) has had to walk out over H, a receiver, in space.

What is the offense hunting, and what’s your call?

Answer 3 is at the end of the chapter. Try it first.

Takeaways

- **Order beats speed.** The master checklist asks nine questions in the order the game answers them (situation, personnel, formation, motion response, box, shell, leverage, pressure, call), and each answer changes what the next clue means.
- **Run or pass is mostly settled by step 3.** In 2024–2025 data the situation and the formation did almost all the work; motion, tempo and the box added under a point. Spend the later steps on the second layer: coverage, rush, where the ball goes.
- **Tells are leans with reasons.** Every tell in the catalogue leaks because of physics or assignment, and every one is broken on purpose by somebody. Trust them in proportion to how often they’re right, and expect the break where it’s cheapest for the offense.
- **Say a number.** A stated confidence between 50% and 90%, made before the snap, turns watching into something you can score. Start from the base rate, move with strong clues, and rarely go past 90.
- **Score the read, not the play.** A correct call on a failed play is a correct call, and a 65% call that misses on a tendency-breaker is honest reading.

Connections

Builds on: every “checklist so far” card (Motion and Shifts, Offensive Systems, Disguise and Rotation, Zone Blitzes); the first tells and the prediction log in How to Watch a Broadcast; tempo and in-game adjustment in In-Game Adjustments and Tempo; the clock in The Clock, the Two-Minute Drill, and Fourth Down; base rates and tendencies in Tendencies: PROE, Personnel, Motion, and Coverage.

Sets up: Film Study: All-22, Charting a Game, and Scoring Yourself, which takes the log you keep live, adds the All-22 film and a full chart of the game, and scores your season of stated confidences with a Brier score and a calibration plot.

Case studies: the film-room drive was run by an offense from the Shanahan family (Shanahan’s Wide Zone Tree) against a defense from the two-high era (Fangio’s Two-High Revolution and Macdonald’s Answer).

i Go deeper

- Atul Gawande, *The Checklist Manifesto* (2009): why short, ordered checklists help experts rather than replace them.
- Philip Tetlock and Dan Gardner, *Superforecasting* (2015): stating probabilities, keeping score and updating, the habits behind this chapter's confidence column.
- Chris B. Brown, *The Art of Smart Football* (2015), and his Smart Football site: the pre-snap game from the play-caller's side.
- Cody Alexander's MatchQuarters newsletter and Ted Nguyen's film breakdowns at *The Athletic*: coverage and pressure tells explained by people who watch the All-22 every week.
- The nflverse project (nflready) and FTN's charting: everything in this chapter's charts, free to download, and the raw material for your own base rates.

Answers to Predict the play

Answer 1.

Run, about 70%; and if it's a pass, expect play-action. *Situation:* first-and-10 is a coin flip (50% passes in 2025). *Personnel and formation:* 21 personnel with the quarterback under center moves it a long way: on first-and-10 from that look, offenses threw only 31% of the time in 2025. *Box:* eight in the box with one safety deep is the defense loading up against the run, and offenses facing it from this look threw a little less often (27%, though that's only 77 snaps).

So why not 85 or 90? Because this is the exact picture play-action is built for. One deep safety and eight defenders committed to the run leave the deep middle and the space behind the linebackers thin, and an offense that gets the linebackers to step up for the fake can throw a post or a deep crossing route behind them (Play-Action, Bootlegs and Screens). *Which way, if it's a run?* The I-formation hides the direction (with the fullback straight behind the quarterback there's no offset to read), so the clue is the box. The safety who walked down did it on the tight end's side, the formation's strength, and he is the eighth man; an offense that sees him usually runs away from him or checks to a pass. Lean left, to the weak side, but only about 55%; plenty of teams run at the strength anyway, with the tight end and fullback as extra blockers. A good log line: *call R, conf 0.70, clue box, l2 "left", 0.55 (if pass, PA shot).*

Answer 2.

Pass, near certain; four rushers is still the best single call (about 60%), and the ball should come out fast. Run or pass is the cheap part: on third-and-4 to 6 from the shotgun with 11 personnel, offenses threw 93% of the time in 2025. Spend the rest on step 8.

Count the deep players first. Two safeties at 12 yards and two corners seven yards off are four defenders who are, for now, playing coverage. That leaves seven for the rush and the underneath zones, and six of them are standing at the line. If all six rushed, there would be one defender underneath, which almost no defense risks with the ball at midfield. Six can come only as Cover 0, man-to-man with nobody deep, and for that both safeties have to walk down; if they're still at 12 when the ball is snapped, six isn't coming. So the realistic menu is: both mugs bail and the four linemen rush (a bluff, with ordinary coverage behind it); one mug comes while a lineman drops into his spot, still four rushers but from unexpected places (a simulated pressure, the creeper of Zone Blitzes); or one mug comes and one drops, a five-man pressure. On third-and-4 to 6 in 2025, defenses sent five or more on only 27% of dropbacks. A pressure look moves that number up, but probably not past half, so "four rush, 60%" is an honest call, and "five or more, 40%" is the alternative you keep in mind.

Where the ball goes: the offense has to protect against six, so the quarterback will likely set the protection to the mugs and have a quick answer built in. Expect a throw in under 2.5 seconds, near the sticks, to the side a dropping linebacker has the farthest to run to. Which side that is, you can't know before the snap: it depends on which mug drops, and that's exactly why the quarterback's answer is a read he makes after the snap. And if a mug drops into the throwing lane, that is the simulated pressure doing its job.

Answer 3.

A lean to pass, about 60%, aimed at the linebacker on the slot. *Situation:* second-and-2 is a run down overall (only 31% passes in 2025), but from the shotgun with 11 personnel it was 57%, and with no huddle 64% (a small sample, 131 snaps, so don't lean on the decimal). Second-and-short is also the down where offenses take shots, because a failure still leaves third-and-short.

Personnel and tempo: this is the substitution freeze at work. Because the offense didn't substitute, the defense had no guaranteed chance to swap a linebacker for a nickel back, so its strongside linebacker is now standing over a slot receiver in space. That is the

matchup the offense sprinted to the line to get: a quick throw or a crossing route at the linebacker, or a deeper shot if the safeties cheat up for the run.

Why only 60%: walking the Sam out leaves the frozen base defense only six in the box, and with the tight end attached the offense has six blockers for them, a fair fight for a run. A team that just gained eight on the ground may simply run it again. Tempo tells you the offense likes something about this picture; it doesn't tell you which of the two things it likes. A good log line: *call P, conf 0.60, clue tempo, l2 "target the slot", 0.55.*